Irrelevant Donors in Causal Panel Methods: Causes and Consequences of Spurious Pre-Treatment Fits

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Assumption 2.1 (Proportional Asymptotics). We consider the high-dimensional regime where the number of donors and pre-treatment periods grow proportionally: $N, T_0 \to \infty$ such that $N/T_0 \to \zeta \in (0, \infty)$. The number of latent factors $r = r_{T_0}$ may grow slowly with $r = o(T_0)$ and $r/T_0 \to 0$.

Assumption 2.2 (Weak Dependence). For all $i \in \{0,1, \dots, N\}$ and $t = 1, \dots, T_0$, the idiosyncratic errors ε_{it} satisfy:

(a) **Weak cross-sectional and serial dependence.** The relevant and irrelevant donor blocks admit well-defined positive-definite limiting contemporaneous covariance matrices $\Omega_{\text{rel}} \succ 0$ and $\Omega_{\text{irrel}} \succ 0$ with eigenvalues uniformly bounded away from zero and infinity: there exist constants $0 < \underline{c} \leq \bar{C} < \infty$ such that

$$\underline{c} \leq \lambda_{\min}(\Omega_{\text{rel}}), \lambda_{\min}(\Omega_{\text{irrel}}) \leq \lambda_{\max}(\Omega_{\text{rel}}), \lambda_{\max}(\Omega_{\text{irrel}}) \leq \bar{C}.$$

The weak cross-sectional dependence extends across blocks: $\mathbb{E}[\varepsilon_{0t}\varepsilon_{it}] = 0$ for all $i \geq 1$, so the treated unit's idiosyncratic error is uncorrelated with all donor idiosyncratic errors.

(b) **Weak serial dependence.** For each i, $\{\varepsilon_{it}\}$ is mean-zero covariance-stationary with absolutely summable autocovariances $\sum_{h=-\infty}^{\infty} |\gamma_i(h)| < \infty$ uniformly in i, and is α-mixing with mixing coefficients satisfying $\alpha_i(h) = O(h^{-\rho})$ for some $\rho > 0$, uniformly in i.

(c) **Long-run variance.** The panel-level long-run variances satisfy $0 < \underline{\omega}^2 \leq \omega_i^2 \leq \bar{\omega}^2 < \infty$ uniformly in i.

(d) **Moment conditions.** There exists $\delta > 0$ such that $\mathbb{E}[|\varepsilon_{it}|^{4+\delta}] \leq C < \infty$ uniformly in i, t.

(e) **Joint factor–idiosyncratic orthogonality and dependence.** For all $i \in \{0,1, \dots, N\}$and all t, s, $\mathbb{E}[f_t \varepsilon_{is}'] = 0$. Moreover, the joint process (f_t, ε_{it}) is α-mixing with coefficients satisfying $\alpha_i(h) = O(h^{-\rho})$ for the same $\rho > 0$ as in Assumption 2.2(b), uniformly in i.

Assumption 2.3 (Regularity of Factors, Loadings, and Tails). The factors f_t are covariance-stationary and ergodic such that $\operatorname{plim}_{T_0 \to \infty} T_0^{-1} F'F = \Sigma_F \succ 0$. The factors f_t and loadings λ_i satisfy $\| f_t \|, \| \lambda_i \| = O_p(1)$ uniformly in t, i. The idiosyncratic error matrices E_{rel} and E_{irrel} admit unit-specific variances $\sigma_i^2 \in [\underline{c}, \bar{c}]$ with $0 < \underline{c} \leq \bar{c} < \infty$ (or, equivalently, the limiting covariances $\Omega_{rel}, \Omega_{irrel}$ defined in Assumption 2.2(a)), and satisfy uniform $(4+\delta)$-moment bounds (or uniformly sub-exponential tails with bounded Orlicz ψ_1-norm).

Proposition 3.1 (Intrinsic Ridge Attenuation – Irreducible Baseline Bias)

To isolate the intrinsic ridge attenuation that arises from any fixed pool of relevant donors, the precise structural vulnerability later masked by irrelevant units we first consider an auxiliary regime in which the donor pool consists solely of a fixed and finite number N_{rel} of relevant units (hence $N_{\text{rel}}/T_0 \to 0$). Let $X_{\text{rel}} = F\Lambda_{\text{rel}}' + E_{\text{rel}}$ satisfy Assumptions 2.2–2.3, where all limits below are understood to hold conditionally on the realized loadings $\{\lambda_i\}_{i=0}^{N_{\text{rel}}}$ (treated as fixed and uniformly bounded, following Assumption 2.3). Then, as $T_0 \to \infty$,

(i) Irreducible bias. For any nonzero treated-unit loading $\lambda_0 \in \mathbb{R}^r$,

$$\frac{1}{T_0} \parallel (I - P_{X_{\text{rel}}})F\lambda_0 \parallel^2 \to_p B_{FP}^2 \triangleq \lambda_0' M \lambda_0 > 0,$$

where

$$M := (\Sigma_F^{-1} + \Lambda_{\text{rel}}' \Omega_{\text{rel}}^{-1} \Lambda_{\text{rel}})^{-1}.$$

(ii) OLS weights. The unconstrained OLS weights satisfy

$$\widehat{w}_{OLS} \to_p G^{-1} \Lambda_{\text{rel}} \Sigma_F \lambda_0,$$

where

$$G := \Lambda_{\text{rel}} \Sigma_F \Lambda_{\text{rel}}' + \Omega_{\text{rel}}.$$

(iii) Factor-space projection. The projection converges on the factor space:

$$T_0^{-1} F' P_{X_{\text{rel}}} F \to_p A_{\text{ridge}} := \Sigma_F \Lambda_{\text{rel}}' G^{-1} \Lambda_{\text{rel}} \Sigma_F.$$

Remark 3.1 (Invariance to factor rotation). The bias B_{FP}^2 and the limiting matrix A_{ridge} depend only on the column spaces of F and Λ_{rel}. They are invariant under simultaneous nonsingular transformations $F \mapsto FH$ and $\Lambda_{\text{rel}} \mapsto \Lambda_{\text{rel}} H^{-1}$ for any invertible $H \in \mathbb{R}^{r \times r}$ (and likewise for $\lambda_0 \mapsto H^{-1}\lambda_0$).

Remark 3.2 (Link to pre-treatment MSE and ATT bias for Unconstrained and Constrained Estimators). Under the factor model, the pre-treatment residuals evaluated at the unconstrained OLS weights, $\widehat{w}_{\text{OLS}}$, satisfy:

$$\frac{1}{T_0} \parallel Y_0^{\text{pre}} - X_{\text{rel}} \widehat{w}_{\text{OLS}} \parallel^2 = \frac{1}{T_0} \parallel (I - P_{X_{\text{rel}}})F\lambda_0 \parallel^2 + \frac{1}{T_0} \parallel \varepsilon_0 \parallel^2 + o_p(1).$$

Let $\sigma_0^2 = \text{plim}\, T_0^{-1} \parallel \varepsilon_0 \parallel^2$ denote the asymptotic contemporaneous variance of the treated unit's idiosyncratic error. The limit $B_{FP}^2 > 0$ implies that the minimized pre-treatment mean-squared error (MSE) in the factor direction remains strictly bounded away from zero as $T_0 \to \infty$. Because the factor model is assumed to hold identically in the post-treatment period with the same loadings λ_0 and factors f_t, the ATT estimation error at each post-treatment period t decomposes as

$$\hat{\tau}_t - \tau_t = (\widehat{w} - \lambda_0)' f_t + (\widehat{w}' \varepsilon_{rel,t} - \varepsilon_{0t}),$$

where $\varepsilon_{rel,t}$ and ε_{0t} are the vectors of donor and treated idiosyncratic errors at time t. The weights $\widehat{w}$ (whether unconstrained OLS or constrained) are measurable with respect to the pre-treatment data and therefore independent of the post-treatment errors. Under the maintained weak dependence, mean-zero, and uniform moment conditions of Assumptions 2.2–2.3 (extended to the post-treatment periods), the idiosyncratic term satisfies $\widehat{w}' \varepsilon_{rel,t} - \varepsilon_{0t} = o_p(1)$ (uniformly in t when $\parallel \widehat{w} \parallel$ is $O_p(1)$, which holds under our assumptions). Therefore,

$$\hat{\tau}_t - \tau_t = (\widehat{w} - \lambda_0)' f_t + o_p(1).$$

The persistent factor-space mismatch $B_{FP}^2 > 0$ thus implies that the ATT bias does not vanish asymptotically for any linear panel estimator whose weights fail to recover λ_0 exactly.

Crucially, this irreducible bias formalizes a strict theoretical lower bound for any restricted linear panel estimator, including strictly constrained Synthetic Control (SC). Let $\mathcal{W}$ denote a restricted weight space (e.g., the unit simplex $\mathcal{W} = \{w \in \mathbb{R}^{N_{\text{rel}}}: w \geq 0, \mathbf{1}'w = 1\}$). By standard projection geometry, the MSE for any constrained estimator $\widehat{w}_{\mathcal{W}} \in \mathcal{W}$ decomposes orthogonally:

$$\frac{1}{T_0} \parallel Y_0^{\text{pre}} - X_{\text{rel}}\widehat{w}_{\mathcal{W}} \parallel^2 = \frac{1}{T_0} \parallel Y_0^{\text{pre}} - X_{\text{rel}}\widehat{w}_{\text{OLS}} \parallel^2 + \frac{1}{T_0} \parallel X_{\text{rel}}(\widehat{w}_{\text{OLS}} - \widehat{w}_{\mathcal{W}}) \parallel^2.$$

Here the orthogonal decomposition follows from the projection geometry of $P_{X_{\text{rel}}} = X_{\text{rel}}(X'_{\text{rel}}X_{\text{rel}})^{+}X'_{\text{rel}}$,which equals the standard projection $X_{\text{rel}}(X'_{\text{rel}}X_{\text{rel}})^{-1}X'_{\text{rel}}$with probability approaching 1 once the sample Gram matrix is nonsingular.

Taking probability limits, the asymptotic pre-treatment MSE of the constrained estimator is strictly bounded below by $B_{FP}^2 + \sigma_0^2 + \Pi_{\mathcal{W}}$, where $\Pi_{\mathcal{W}} \geq 0$ is the non-negative asymptotic penalty induced by the boundary constraints.

Thus, restricting the weight space cannot resolve the attenuation bias and may raise the lower bound on the (non-vanishing) asymptotic ATT bias if $\widehat{w}_{OLS} \notin \mathcal{W}$. The fundamental inability of a fixed relevant-donor pool to achieve zero pre-treatment MSE, whether unconstrained or constrained, creates the precise empirical vulnerability that is later masked by spurious interpolation when irrelevant donors are introduced to the pool.

Remark 3.3 (General scaling of the ridge matrix). The ridge matrix $M = (\Sigma_F^{-1} + \Lambda'_{\text{rel}}\Omega_{\text{rel}}^{-1}\Lambda_{\text{rel}})^{-1}$ satisfies the general scaling $M = O_p(N_{\text{rel}}^{-1})$ under the pervasiveness condition $\lambda_{\min}(N_{\text{rel}}^{-1}\Lambda'_{\text{rel}}\Omega_{\text{rel}}^{-1}\Lambda_{\text{rel}}) \geq c > 0$ (already invoked in the proof of Proposition 3.1 and stated explicitly in Corollary 3.1). Consequently, when the number of relevant donors is held fixed and finite as in the present proposition, we have $M = \Theta(1)$ and the irreducible bias satisfies $B_{\text{FP}}^2 = \lambda_0' M \lambda_0 = \Theta(1) > 0$ for any nonzero treated-unit loading λ_0. When $N_{\text{rel}} \to \infty$ under the pervasiveness condition, $M \to 0$ deterministically (since M is a deterministic function of the fixed loading and covariance parameters); consequently $B_{\text{FP}}^2 = \lambda_0' M \lambda_0 \to 0$ is an ordinary limit, not merely convergence in probability. Thus the fixed-N_{rel} regime serves as the leading illustration of insufficient relevance: even with perfect spanning capability ($N_{\text{rel}} \geq r$ and Λ_{rel} full column rank), the non-vanishing noise covariance Ω_{rel} generates a permanent generalized ridge attenuation on the factor space. The bias vanishes precisely when the relevant donor pool itself grows (Corollary 3.1).

Interpretation. We don't emphasize the condition $N_{rel} < r$, once its result is a trivial spanning bias given that the relevant pool structurally cannot span the latent factors. The core insight of Proposition 3.1, however, is that even when the donor pool has a fixed number of relevant donors with perfect spanning capability ($N_{rel} \geq r$ and Λ_{rel} has full column rank), the surviving noise covariance Ω_{rel} induces a permanent shrinkage. This intrinsic ridge attenuation is explicitly manifested in the probability limit of the unconstrained weights, plim $\widehat{w}_{OLS} = (\Lambda_{rel}\Sigma_F\Lambda'_{rel} + \Omega_{rel})^{-1}\Lambda_{rel}\Sigma_F\lambda_0$, demonstrating exactly how the surviving noise covariance Ω_{rel} acts as a permanent, structural L_2-style penalty that shrinks the estimator away from the ideal latent factor loadings. The unconstrained OLS projection therefore converges (on the factor space) to the distorted operator characterized by A_{ridge} that systematically under-matches the treated unit's factor component, generating the strictly positive baseline bias $B_{FP}^2 > 0$ even in infinite samples. This fundamental limitation of any relevant-donor pool is

the point of vulnerability that is later masked by spurious interpolation once irrelevant donors are added.

Relation to existing literature. The irreducible baseline bias established in Proposition 3.1 formally connects to and extends known impossibility results in the Synthetic Control (SC) literature. Most notably, Ferman and Pinto (2021) demonstrate that when the donor pool is fixed, SC weights generally fail to converge to the true data-generating factor loadings even as $T_0 \to \infty$, because the variance of the idiosyncratic noise does not vanish from the asymptotic objective function. Proposition 3.1 complements this insight by providing an explicit algebraic characterization of this asymptotic distortion. This irreducible baseline bias is precisely the multivariate generalization of classical errors-in-variables attenuation, in which the donor matrix X_{rel} serves as a noisy proxy for the latent factor component; the Woodbury-derived ridge matrix M isolates the exact algebraic mechanism of the permanent shrinkage and thereby supplies the clean foundation for showing how irrelevant donors subsequently mask this structural deficit through spurious interpolation. By applying the Woodbury identity to the limiting Gram matrix in the unconstrained space, we isolate the exact mechanism of the bias: the surviving noise covariance Ω_{rel} acts as a permanent, generalized ridge penalty on the latent factor space. This formulation is highly analytically tractable, directly bridging the unconstrained attenuation to the restricted SC geometry in Remark 3.2, and precisely isolating the pervasiveness condition required for the bias to vanish as the relevant donor pool grows (Corollary 3.1). Crucially, establishing this baseline vulnerability serves as the necessary foundation to show how the subsequent introduction of irrelevant donors destructively masks this structural deficit via spurious interpolation.

Proof.

Claim (i): Bias limit. All limits are as $T_0 \to \infty$ with fixed N_{rel}, and hold in probability.

The treated unit's pre-treatment vector is given by the factor model

$$\mathrm{Y}_0^{\mathrm{pre}} = \mathrm{F}\lambda_0 + \varepsilon_0.$$

The linear projection of $\mathrm{Y}_0^{\mathrm{pre}}$ onto the column space of the donor matrix $\mathrm{X}_{\mathrm{rel}}$ is

$$\mathrm{P}_{\mathrm{X}_{\mathrm{rel}}}\mathrm{Y}_0^{\mathrm{pre}},$$

The orthogonal projection matrix is $P_{X_{\mathrm{rel}}} = X_{\mathrm{rel}}(X_{\mathrm{rel}}'X_{\mathrm{rel}})^{+}X_{\mathrm{rel}}'$,where $+$ denotes the Moore–Penrose pseudoinverse. Since the limiting Gram matrix $G = \Lambda_{\mathrm{rel}}\Sigma_F\Lambda_{\mathrm{rel}}' + \Omega_{\mathrm{rel}} \succ 0$(established below), the sample Gram $T_0^{-1}X_{\mathrm{rel}}'X_{\mathrm{rel}}$is nonsingular with probability approaching 1, so $(X_{\mathrm{rel}}'X_{\mathrm{rel}})^{+} = (X_{\mathrm{rel}}'X_{\mathrm{rel}})^{-1}$eventually. The pseudoinverse formulation avoids conditioning on finite-sample invertibility events and is adopted throughout the proof. The projection mismatch for the treated unit is therefore

$$\mathrm{Y}_0^{\mathrm{pre}} - \mathrm{P}_{\mathrm{X}_{\mathrm{rel}}}\mathrm{Y}_0^{\mathrm{pre}} = (\mathrm{I} - \mathrm{P}_{\mathrm{X}_{\mathrm{rel}}})\mathrm{Y}_0^{\mathrm{pre}} = (\mathrm{I} - \mathrm{P}_{\mathrm{X}_{\mathrm{rel}}})(\mathrm{F}\lambda_0 + \varepsilon_0).$$

Under the maintained assumptions, the idiosyncratic component ε_0 is asymptotically orthogonal to the factor space, so the irreducible bias of interest is fully captured by the mismatch on the latent factor component:

$$(\mathrm{I} - \mathrm{P}_{\mathrm{X}_{\mathrm{rel}}})\mathrm{F}\lambda_0.$$

Equation 1 (exact finite-sample decomposition of the mismatch). Let $\mathrm{B_{rel}} := (\mathrm{I} - \mathrm{P_{X_{rel}}})\mathrm{F}\lambda_0$. Because $\mathrm{I} - \mathrm{P_{X_{rel}}}$ is symmetric and idempotent, the normalized squared norm admits the exact finite-sample decomposition

$$\frac{1}{T_0} \| \mathrm{B_{rel}} \|^2 = \underbrace{\lambda_0'(\frac{\mathrm{F'F}}{T_0})\lambda_0}_{\substack{\breve{Term}\ A: \\ Treated\ factor \\ quadratic\ form}} - \lambda_0'(\frac{\mathrm{F'X_{rel}}}{T_0})(\underbrace{\frac{\mathrm{X_{rel}'X_{rel}}}{T_0}}_{\substack{\breve{Term}\ G: \\ Full\ donor \\ Gram\ matrix}})^{+}(\frac{\mathrm{X_{rel}'F}}{T_0})\lambda_0.$$

Equation 2 (full expansion of the donor Gram matrix)

Expand $\mathrm{X_{rel}} = \mathrm{F}\Lambda_{\mathrm{rel}}' + \mathrm{E_{rel}}$. The donor Gram decomposes as

$$\frac{1}{T_0}\mathrm{X_{rel}'X_{rel}} = \underbrace{\Lambda_{\mathrm{rel}}(\frac{\mathrm{F'F}}{T_0})\Lambda_{\mathrm{rel}}'}_{\substack{\breve{Term}\ S: \\ Signal\ term}} + \underbrace{\Lambda_{\mathrm{rel}}(\frac{\mathrm{F'E_{rel}}}{T_0})}_{\substack{\breve{Term}\ C1: \\ First\ cross\ term}} + \underbrace{(\frac{\mathrm{E_{rel}'F}}{T_0})\Lambda_{\mathrm{rel}}'}_{\substack{\breve{Term}\ C2: \\ Second\ cross\ term}} + \underbrace{\frac{1}{T_0}\mathrm{E_{rel}'E_{rel}}}_{\substack{\breve{Term}\ N: \\ Noise\ term}}.$$

We now establish the convergence of each labeled term in turn, using the unified concentration results under α-mixing with rate $\rho > 1$, uniform $(4+\delta)$-moments, and fixed dimension.

Convergence of Term A (treated factor quadratic form) By covariance-stationarity and ergodicity of the factors (Assumption 2.3),

$$\frac{1}{T_0}\mathrm{F'F} = \Sigma_F + o_p(1).$$

Convergence of Term N (noise term) The noise Gram is $\frac{1}{T_0}\mathrm{E_{rel}'E_{rel}}$. Under the maintained weak dependence and moment conditions of Assumption 2.2 together with the joint orthogonality and mixing conditions in Assumption 2.3(e), the same concentration results yield

$$\| \frac{1}{T_0}\mathrm{E_{rel}'E_{rel}} - \Omega_{\mathrm{rel}} \|_{\mathrm{op}} = O_p(\sqrt{\frac{N_{\mathrm{rel}}}{T_0}}).$$

Here Ω_{rel} denotes the contemporaneous covariance matrix of the vector of relevant donor idiosyncratic errors (i.e., $p\lim_{T_0\to\infty} T_0^{-1}\mathrm{E_{rel}'E_{rel}}$).

With N_{rel} is fixed, the rate simplifies to $O_p(1/\sqrt{T_0})$. As $T_0 \to \infty$, $O_p(1/\sqrt{T_0}) \to 0$, and therefore

$$\frac{1}{T_0}\mathrm{E_{rel}'E_{rel}} \to_p \Omega_{\mathrm{rel}}.$$

Convergence of Terms C1 and C2 (cross terms) The cross terms are $\frac{1}{T_0}\Lambda_{\mathrm{rel}}'\mathrm{F'E_{rel}}$ and its transpose. Under the joint orthogonality and mixing conditions in Assumption 2.3(e), the product process $\{f_t\varepsilon_{it}'\}$ is itself α-mixing with rate ρ. Since $\| f_t \| = O_p(1)$ uniformly in t (Assumption 2.3) and the idiosyncratic errors admit uniform (4+δ)-moments (Assumption 2.2(d)), Hölder's inequality implies that the product process satisfies the same uniform moment bounds required by the cited concentration results. The concentration results for sample covariances of α-mixing vector processes (Merlevède, Peligrand and Rio, 2011; Banna, Merlevède and Youssef, 2016) therefore yield

$$\| \frac{1}{T_0} \Lambda'_{\mathrm{rel}} \mathrm{F}' \mathrm{E}_{\mathrm{rel}} \|_{\mathrm{op}} = O_p({T_0}^{-1/2}).$$

Since N_{rel} is fixed, this is a bounded constant. As $T_0 \to \infty$, $O_p({T_0}^{-1/2}) \to 0$, and therefore

$$\frac{1}{T_0} \mathrm{F}' \mathrm{E}_{\mathrm{rel}} \ \to_p \ 0.$$

Convergence of Term S (signal term) Term S is $\Lambda_{\mathrm{rel}}(\frac{\mathrm{F}'\mathrm{F}}{T_0})\Lambda'_{\mathrm{rel}}$. Substituting the limit of Term A gives

$$\Lambda_{\mathrm{rel}}(\frac{\mathrm{F}'\mathrm{F}}{T_0})\Lambda'_{\mathrm{rel}} \ \to_p \ \Lambda_{\mathrm{rel}} \Sigma_F \Lambda'_{\mathrm{rel}}.$$

Convergence of Term G (the full donor Gram matrix - Equation 2). Substituting the expansions and the three limits above yields

$$\frac{1}{T_0} X'_{\mathrm{rel}} X_{\mathrm{rel}} = \Lambda_{\mathrm{rel}}(\frac{1}{T_0} F'F)\Lambda'_{\mathrm{rel}} + \text{cross terms (rate } O_p(T_0^{-1/2})) + \frac{1}{T_0} E'_{\mathrm{rel}} E_{\mathrm{rel}}.$$

Taking the limit,

$$\frac{1}{T_0} X'_{\mathrm{rel}} X_{\mathrm{rel}} \to_p G := \Lambda_{\mathrm{rel}} \Sigma_F \Lambda'_{\mathrm{rel}} + \underbrace{\Omega_{\mathrm{rel}}}_{\text{ridge penalty}}.$$

Here Ω_{rel} is the source of the ridge penalty: it is the permanent, non-vanishing noise covariance of the relevant donors that will act as a structural L_2-regularization term once the Woodbury identity is applied (see below). Because $\Omega_{\mathrm{rel}} \succ 0$ with eigenvalues bounded away from zero (Assumption 2.2(a)) and $\Sigma_F \succ 0$ (Assumption 2.3), the deterministic limiting matrix G is strictly positive definite and invertible. Consequently, the sample Gram matrix $T_0^{-1} X'_{\mathrm{rel}} X_{\mathrm{rel}}$ is invertible with probability approaching 1. Finally,

$$\frac{1}{T_0} F' X_{\mathrm{rel}} = \frac{1}{T_0} F'F\Lambda'_{\mathrm{rel}} + \frac{1}{T_0} F' E_{\mathrm{rel}},$$

therefore

$$\frac{1}{T_0} F' X_{\mathrm{rel}} \to_p \Sigma_F \Lambda'_{\mathrm{rel}}.$$

Limit of the quadratic form. Inserting the four limits into the exact decomposition gives

$$\frac{1}{T_0} \| \mathrm{B}_{\mathrm{rel}} \|^2 \to_p \lambda'_0 \Sigma_F \lambda_0 - \lambda'_0 (\Sigma_F \Lambda'_{\mathrm{rel}}) \mathrm{G}^{-1} (\Lambda_{\mathrm{rel}} \Sigma_F) \lambda_0 = \lambda'_0 [\Sigma_F - \Sigma_F \Lambda'_{\mathrm{rel}} \mathrm{G}^{-1} \Lambda_{\mathrm{rel}} \Sigma_F] \lambda_0.$$

Woodbury simplification. Apply the Woodbury matrix identity with $\mathrm{A} = \Sigma_F$, $\mathrm{B} = \Lambda_{\mathrm{rel}}$, $\mathrm{C} = \Omega_{\mathrm{rel}}$:

$$\mathrm{A} - \mathrm{AB}'(\mathrm{BAB}' + \mathrm{C})^{-1} \mathrm{BA} = (\mathrm{A}^{-1} + \mathrm{B}'\mathrm{C}^{-1}\mathrm{B})^{-1}.$$

Hence

$$\Sigma_F - \Sigma_F \Lambda'_{\mathrm{rel}} \mathrm{G}^{-1} \Lambda_{\mathrm{rel}} \Sigma_F = (\Sigma_F^{-1} + \underbrace{\Lambda_{\mathrm{rel}}^{\top} \Omega_{\mathrm{rel}}^{-1} \Lambda_{\mathrm{rel}}}_{\text{ridge term}})^{-1} \triangleq \mathrm{M}.$$

Explicit scaling of M. From the Woodbury matrix identity established above, we have

$$\mathrm{M} = (\Sigma_F^{-1} + \Lambda_{\mathrm{rel}}^{\top}\Omega_{\mathrm{rel}}^{-1}\Lambda_{\mathrm{rel}})^{-1}.$$

In general, the ridge term satisfies $\Lambda_{\mathrm{rel}}^{\top}\Omega_{\mathrm{rel}}^{-1}\Lambda_{\mathrm{rel}} = \Theta(N_{\mathrm{rel}})$ (by the eigenvalue bounds in Assumption 2.2(a) when N_{rel} is fixed, or by the pervasiveness condition $\lambda_{\min}(N_{\mathrm{rel}}^{-1}\Lambda_{\mathrm{rel}}^{\top}\Omega_{\mathrm{rel}}^{-1}\Lambda_{\mathrm{rel}}) \geq c > 0$ together with Weyl's inequality applied to the sample Gram matrix $T_0^{-1}\mathrm{X}_{\mathrm{rel}}'\mathrm{X}_{\mathrm{rel}}$ when N_{rel} grows; see the argument in the proof of Corollary 3.1). It follows that

$$\mathrm{M} = O_p(N_{\mathrm{rel}}^{-1}).$$

In the present proposition, where the number of relevant donors N_{rel} is held fixed and finite, the general scaling reduces to

$$\mathrm{M} = O_p(N_{\mathrm{rel}}^{-1}) = \Theta(1) > 0.$$

Consequently, $B_{FP}^2 = \lambda_0'\mathrm{M}\lambda_0 = \Theta(1) > 0$ for any nonzero treated-unit loading λ_0. This holds unconditionally, even if $N_{\mathrm{rel}} < r$. □

Connection to Corollary 3.1. The proof above retains exactly the same general expressions and rates as will be used in Corollary 3.1: the factor Gram rate $o_p(1)$, the noise Gram rate $O_p(\sqrt{N_{\mathrm{rel}}/T_0})$, the cross term controlled by the same concentration apparatus, and the general scaling $\mathrm{M} = O_p(N_{\mathrm{rel}}^{-1})$.

Claim (ii): OLS weight limit. The OLS weights satisfy $\widehat{w}_{OLS} = (X_{\mathrm{rel}}'X_{\mathrm{rel}})^{+}X_{\mathrm{rel}}'Y_0^{\mathrm{pre}}$. We analyze the normalized version

$$T_0^{-1}X_{\mathrm{rel}}'Y_0^{\mathrm{pre}} = T_0^{-1}X_{\mathrm{rel}}'F\lambda_0 + T_0^{-1}X_{\mathrm{rel}}'\varepsilon_0.$$

The first term satisfies

$$T_0^{-1}X_{\mathrm{rel}}'F = \Lambda_{\mathrm{rel}}(T_0^{-1}F'F) + T_0^{-1}E_{\mathrm{rel}}'F \to_p \Lambda_{\mathrm{rel}}\Sigma_F,$$

using the factor Gram limit from Claim (i) and the convergence of the cross term $T_0^{-1}F'E_{\mathrm{rel}} \to_p 0$ established under Assumption 2.3(e).

For the second term, expand:

$$T_0^{-1}X_{\mathrm{rel}}'\varepsilon_0 = \Lambda_{\mathrm{rel}}(T_0^{-1}F'\varepsilon_0) + T_0^{-1}E_{\mathrm{rel}}'\varepsilon_0.$$

Under the joint orthogonality and mixing conditions in Assumption 2.3(e), the process $\{f_t\varepsilon_{0t}'\}$ is α-mixing, so $T_0^{-1}F'\varepsilon_0 \to_p 0$ by the mixing weak law of large numbers. For $T_0^{-1}E_{\mathrm{rel}}'\varepsilon_0$, each entry is a sample average of mean-zero terms (by the cross-sectional uncorrelatedness condition in Assumption 2.2(a)). Under Assumption 2.3(e) and the concentration results for α-mixing processes (Merlevède, Peligrand and Rio, 2011; Banna, Merlevède and Youssef, 2016), we obtain $T_0^{-1}E_{\mathrm{rel}}'\varepsilon_0 \to_p 0$.

Hence $T_0^{-1}X_{\mathrm{rel}}'\varepsilon_0 \to_p 0$. Combining this with $T_0^{-1}X_{\mathrm{rel}}'X_{\mathrm{rel}} \to_p G$ from Claim (i) and applying the continuous mapping theorem yields

$$\widehat{w}_{OLS} \to_p G^{-1}\Lambda_{\mathrm{rel}}\Sigma_F\lambda_0. \square$$

Claim (iii): Factor-space projection. Using the same limits:

$$T_0^{-1}F'P_{X_{\text{rel}}}F = (T_0^{-1}F'X_{\text{rel}})(T_0^{-1}X'_{\text{rel}}X_{\text{rel}})^{-1}(T_0^{-1}X'_{\text{rel}}F).$$

By the continuous mapping theorem:

$$T_0^{-1}F'P_{X_{\text{rel}}}F \to_p (\Sigma_F\Lambda'_{\text{rel}})G^{-1}(\Lambda_{\text{rel}}\Sigma_F) = A_{\text{ridge}}. \square$$

Corollary 3.1 (Baseline Bias Vanishes with Growth of Relevant Donors)

Suppose additionally that the relevant-donor loadings satisfy the strong pervasiveness condition $\lambda_{\min}(N_{\text{rel}}^{-1}\Lambda'_{\text{rel}}\Omega_{\text{rel}}^{-1}\Lambda_{\text{rel}}) \geq c > 0$ uniformly in N_{rel}. Assume further that the mixing coefficients in Assumption 2.2(b) satisfy $\rho > 1$. If $N_{rel} \to \infty$ such that $N_{rel}/T_0 \to \zeta_{rel} \in [0,1)$, allowing for both sublinear and proportional growth of the relevant donor pool, the baseline bias vanishes: $B_{FP}^2 \to 0$ (see Remark 3.3 for the complementary scaling when N_{rel} is fixed).

Interpretation: This clarifies the fundamental tradeoff, a fixed relevant donor pool leaves an irreducible baseline attenuation due to un-averaged idiosyncratic noise. However, as the relevant pool grows (even proportionally to T_0), the latent factors perfectly separate from the bounded noise, eliminating the structural bias. Note that this convergence is deterministic (not merely in probability): $B_{\text{FP}}^2 = \lambda_0' M \lambda_0$is a deterministic functional of the fixed loading matrices and covariance parameters, so the limit as $N_{\text{rel}} \to \infty$is an ordinary limit of a real sequence. This establishes the vital theoretical baseline: adding relevant donors cures the baseline bias, but as we show in the remainder of the paper, opening the donor pool to irrelevant units destructively masks this vulnerability via spurious interpolation.